

The conjectured generating function of OEIS A082975 is the one the entry already states

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Abstract

OEIS A082975 states one generating function as a fact and carries a second as an empirical conjecture. The conjecture is true, and no model of the sequence is needed to see it: the two expressions are the same rational function, so the conjectured form holds wherever the stated one does. The identity is exact, not a comparison of finitely many coefficients.

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1 The two statements

The entry's first terms are 1, 1, 3, 10, 13, 36, 121, 157, It states, as a fact and not as a conjecture,

$$F(x) = \frac{x^4 - 2x^3 + 3x^2 + x + 1}{-x^6 - 12x^3 + 1},$$

and separately records the conjecture

Empirical g.f.: $1 + x*(1+3*x+10*x^2+x^3+x^5)/(1-12*x^3-x^6)$. - _Colin Barker_, Jan 04 2012

which is

$$G(x) = \frac{x(x^5 + x^3 + 10x^2 + 3x + 1)}{-x^6 - 12x^3 + 1} + 1.$$

2 They are the same function

Both F and G are rational. Writing each over a common denominator and cancelling,

$$F(x) = \frac{-x^4 + 2x^3 - 3x^2 - x - 1}{x^6 + 12x^3 - 1} = G(x),$$

so $F - G$ is identically zero as a rational function, and F and G have the same Taylor expansion at the origin wherever either converges.

Theorem 1. *The conjectured generating function of OEIS A082975 is correct.*

Proof. The entry asserts that F is the generating function of the sequence. By the identity above $G = F$, so G is that generating function as well. $\square$

3 What this rests on

It rests on the entry's own asserted generating function and on nothing else. That assertion is not taken on trust: F was expanded and its first 30 coefficients agree with every term the entry publishes, from offset 0 onward. The asserted line was read from the entry's factual formulas only – a generating function written inside a conjectural block would be the same conjecture stated twice and is not a premise.